

BRCA1 ANALYSIS REPORT

Project Title:

**Comprehensive Bioinformatics Analysis of the Human
BRCA1 Gene Using NCBI Databases**

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Project Number: Project 03

Domain: Bioinformatics | Clinical Genomics

Abstract

The BRCA1 (Breast Cancer Type 1 Susceptibility Gene) is a well-known tumor suppressor gene that plays a critical role in maintaining genome stability through DNA repair mechanisms. Mutations in this gene are strongly associated with hereditary breast and ovarian cancers. This project presents a comprehensive bioinformatics analysis of the human BRCA1 gene using publicly available databases provided by the National Center for Biotechnology Information (NCBI).

The analysis involved retrieving gene, transcript, nucleotide, and protein information from NCBI databases, studying the genomic organization of BRCA1 using the Genome Data Viewer, investigating clinically significant variants through ClinVar, validating the nucleotide sequence using the BLAST tool, and exploring the biological functions of BRCA1 using Gene Ontology (GO) annotations.

The BLAST analysis confirmed a 100% query coverage, 100% sequence identity, and an E-value of 0.0, indicating a perfect match with the reference BRCA1 sequence. Functional annotation demonstrated the involvement of BRCA1 in DNA repair, cell cycle regulation, and maintenance of genomic integrity.

This project provided practical experience in navigating major bioinformatics databases, interpreting genomic and proteomic information, understanding clinical variant databases, and documenting scientific findings in a professional format. The skills acquired through this project establish a strong foundation for advanced studies in bioinformatics, next-generation sequencing (NGS), and clinical genomics.

Keywords: BRCA1, Bioinformatics, NCBI, ClinVar, BLAST, Gene Ontology, Clinical Genomics, DNA Repair, Genome Analysis.

Tools Used:

- NCBI Gene
- NCBI Nucleotide
- NCBI Protein
- ClinVar
- BLAST
- Gene Ontology

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1. Introduction

BRCA1 (Breast Cancer Type 1 Susceptibility Gene) is a tumor suppressor gene located on chromosome 17. It plays an essential role in DNA repair, genome stability, and cell cycle regulation. Mutations in BRCA1 are strongly associated with hereditary breast and ovarian cancers.

This project explores the BRCA1 gene using publicly available bioinformatics databases to understand its sequence, structure, function, and clinical significance.

2. Objective

The objectives of this project are:

1. To identify the BRCA1 gene using NCBI Gene.
2. To study its transcript and protein information.
3. To analyze the coding sequence (CDS).
4. To understand the genomic organization of BRCA1.
5. To investigate clinically significant variants using ClinVar.
6. To validate the BRCA1 sequence using BLAST.
7. To study the biological functions of BRCA1 using Gene Ontology.

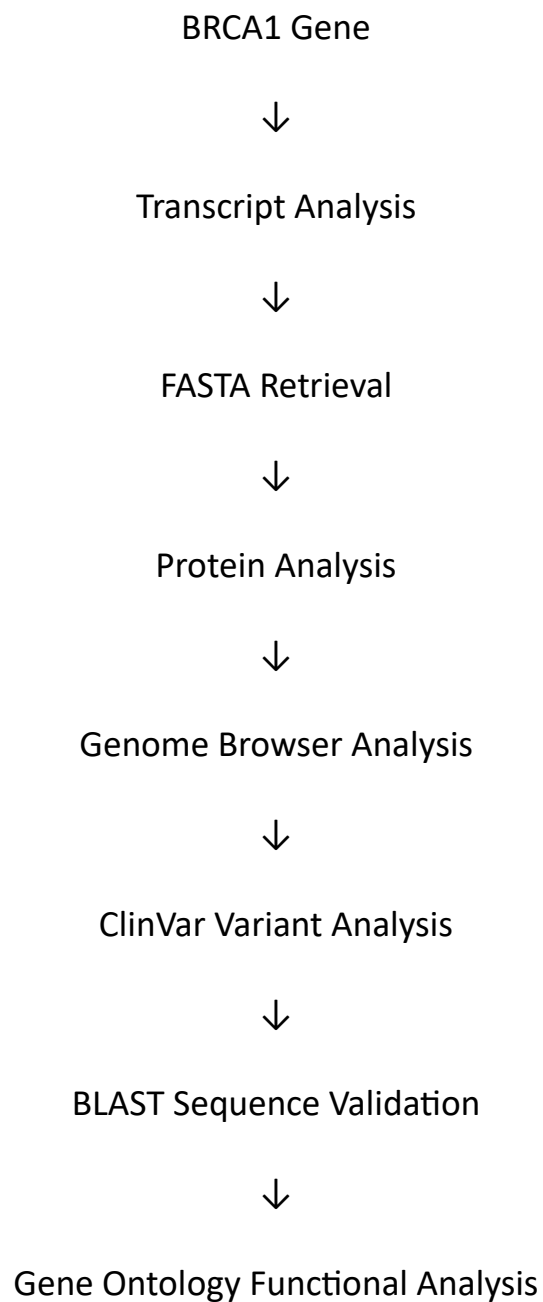
3. Databases Used

Database	Purpose
NCBI gene	Gene information
NCBI Nucleotide	Transcript & FASTA
NCBI Protein	Protein Sequence

ClinVar	Clinical Variant
BLAST	Sequence Validation
Gene Oncology	Functional Annotation

4. Methodology

The following workflow was used during the analysis:



5. Results

5.1 Gene Information

Gene: BRCA1

Organism: Homo sapiens

Chromosome: 17

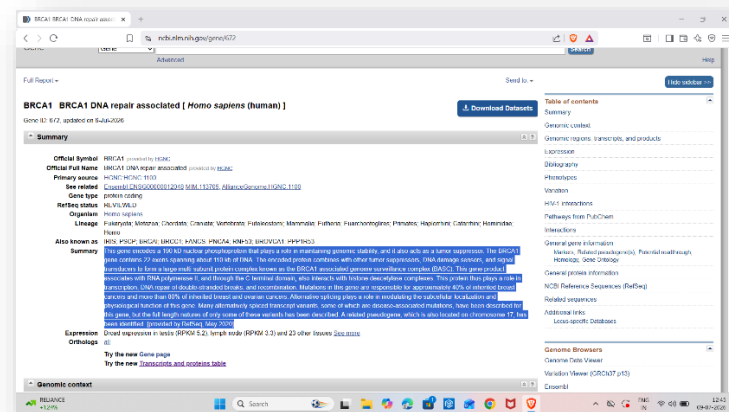


Figure 1: Overview of BRCA1 gene from the NCBI Gene database showing gene name, chromosome location and number

5.2 Transcript Analysis

- ✓ MANE Select transcript identified.
- ✓ Transcript accession recorded.
- ✓ FASTA sequence downloaded.

Protein Name	Accession Number	Length
atp1a1	F.000000000	1863
atp1a2	F.000000001	1863

5.3 Protein Analysis

- Protein accession identified.
- Protein length: 1863 amino acids

breast cancer type 1 susceptibility protein isoform 1 [Homo sapiens]

NCBI Reference Sequence: NP_009225.1

[Identical Proteins](#) [FASTA](#) [Graphics](#)

Go to: [▼](#)

LOCUS **NP_009225** **1863 aa** linear PRI 04-JUN-2026

DEFINITION breast cancer type 1 susceptibility protein isoform 1 [Homo sapiens].

ACCESSION **NP_009225**

VERSION **NP_009225.1**

DBSOURCE REFSEQ: accession [NM_007294.4](#)

KEYWORDS RefSeq; MANE Select.

SOURCE Homo sapiens (human)

ORGANISM **Homo sapiens**
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo.

REFERENCE 1 (residues 1 to 1863)
Orban, T.I. and Olah, E.

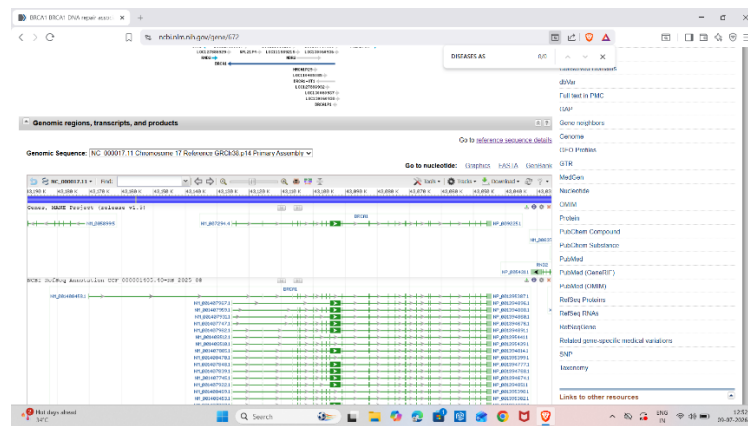
TITLE Emerging roles of BRCA1 alternative splicing

5.4 Genome Analysis

The Genome Data Viewer was used to visualize:

- Gene location

- Exons
- Introns
- Chromosomal structure



5.5 ClinVar Analysis

ClinVar revealed multiple BRCA1 variants associated with hereditary breast and ovarian cancer.

Concepts studied:

- Pathogenic variants
- Benign variants
- HGVS nomenclature
- Copy Number Variation (CNV)

Global minor allele frequency (gnaf)

Allele frequency

Links: dbVar: nsv1415284 dbVar: nsv817369 VarSome

Genes

Gene	OMIM	ClinGen Gene Dosage Sensitivity Curation	TS score	Variation Viewer	Related variants
AXIN2		Sufficient evidence for dosage pathogenicity	No evidence available	GRCh38 GRCh37	4320 4334
BPTF		Sufficient evidence for dosage pathogenicity	No evidence available	GRCh38 GRCh37	1158 1222
BRCA1		Sufficient evidence for dosage pathogenicity	No evidence available	GRCh38 GRCh37	13980 15986
BRP1		Sufficient evidence for dosage pathogenicity	No evidence available	GRCh38 GRCh37	6472 6530
CLTC		Sufficient evidence for dosage pathogenicity	No evidence available	GRCh38 GRCh37	1074 1258
COL1A1		Sufficient evidence for dosage pathogenicity	No evidence available	GRCh38 GRCh37	3596 3864
EFTUD2		Sufficient evidence for dosage pathogenicity	No evidence available	GRCh38 GRCh37	959 976

NOG

There are 2029 more genes affected by this variant. See the full set of genes in Variation Viewer (GRCh38, GRCh37, NCBI36) and ClinGen Dosage Sensitivity Map.

Conditions - Germline

Condition	Classification (# of submissions)	Review status	Last evaluated	Variation/condition record
See cases	Pathogenic (1)	★★★★	Apr 21, 2011	RCV000137437.10

Submissions - Germline

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows
			ISCA site 1 Additional submitter: International Standards For Cytogenomic Arrays Consortium (ISCA) Accession: SCV000177658.4 First in ClinVar: Aug 30, 2014 Last updated: Oct 26, 2017 Comments (2) Copy number variation identified through the course of routine clinical cytogenomic testing in nonrural populations with clinical	

ClinVar Home About Understand the Data Access the Data Submit Data

Graphical view of search results

Download Select Columns Sort by: Location

Search results: 15,986 variants for BRCA1[gene]

Variation	Gene	Type	Condition	Classification, Review status
GRCh38:hg38.17:621,231,109,1738,499,226,735,931,934,939	AARS01, AKT1	copy number gain	See cases	Pathogenic ★
GRCh38:hg38.17:621,332,242,091,738,499,226,581,702,146,3	AARS01, AKT1	copy number gain	See cases	Pathogenic ★

5.6 BLAST Analysis

Sequence validation produced:

- Query Coverage: 100%
- Identity: 100%
- E-value: 0.0

Interpretation:

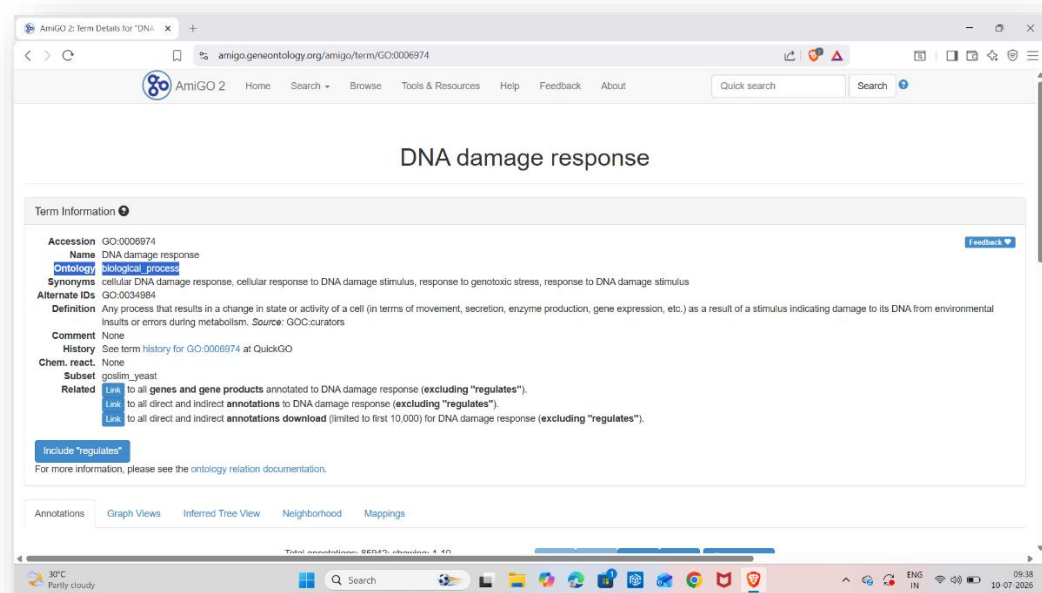
The uploaded sequence perfectly matched the human BRCA1 reference sequence.

The screenshot displays the NCBI BLAST results page for a BLASTN search. The browser address bar shows the URL `blast.ncbi.nlm.nih.gov/Blast.cgi`. The page header includes the NIH logo and the text "National Library of Medicine National Center for Biotechnology Information". The main heading is "BLAST® » blastn suite » results for RID-516J8TX1014". Below this, there is a banner for "Join the BLAST blastn newblastresults testing community" with an "Email us" button. The search parameters are listed on the left: Job Title "NC_000017.11:c43170327-43044295 Homo sapiens...", RID "516J8TX1014", Program "BLASTN", Database "core_nt", Query ID "IcIQuery_4029947", Description "NC_000017.11:c43170327-43044295 Homo sapiens chromoso...", Molecule type "dna", and Query Length "7236". On the right, the "Filter Results" section allows filtering by Organism, Percent Identity, E value, and Query Coverage. The "Sequences producing significant alignments" section is at the bottom, showing a table with columns for "select all", "0 sequences selected", "GenBank", "Graphics", "Distance tree of results", and "MSA Viewer". The Windows taskbar at the bottom shows the date and time as "12:38 10-07-2025".

5.7 Gene Ontology

1. Biological Process

- DNA repair
- Cell cycle regulation
- DNA damage response



2. Molecular Function

- DNA binding
- Protein binding
- Ubiquitin ligase activity

BRCA1 BRCA1 DNA repair assoc

ncbi.nlm.nih.gov/gene/672

Homology

NCBI Orthologs

Orthologs from OrthoDB

Gene Ontology Provided by GOA

Items 1 - 25 of 29 < Prev

Function	Evidence
enables DNA binding	IEA
enables DNA binding	TAS
enables RNA binding	IDA
enables RNA polymerase binding	IDA
enables damaged DNA binding	IBA
enables damaged DNA binding	IEA
enables enzyme binding	IPI
enables histone H2AK127 ubiquitin ligase activity	TAS
enables histone H2AK129 ubiquitin ligase activity	TAS
enables identical protein binding	IPI

30°C Partly cloudy

3. Cellular Component

- Nucleus
- DNA repair complex

AmiGO 2 Term Details for "BRCA1-A complex"

amigo.geneontology.org/amigo/term/GO:0070531

AmiGO 2 Home Search Browse Tools & Resources Help Feedback About

Quick search Search

BRCA1-A complex

Term Information

Accession GO:0070531

Name BRCA1-A complex

Ontology cellular_component

Synonyms None

Alternate IDs None

Definition A protein complex that contains the BRCA1-BARD1 heterodimer, RAP80/UIMC1, BRCC3/BRCC36, BRE/BRCC45, FAM175A/CCDC38/Abraxas and MERIT40/INBA1, and specifically recognizes and binds K63-linked polyubiquitin chains present on histone H2A and H2AX at DNA damage sites. Source: GOC:mah, PMID:19261749

Comment None

History See term history for GO:0070531 at QuickGO

Chem. react. None

Subset None

Related

- [Link](#) to all genes and gene products annotated to BRCA1-A complex (excluding "regulates").
- [Link](#) to all direct and indirect annotations to BRCA1-A complex (excluding "regulates").
- [Link](#) to all direct and indirect annotations **download** (limited to first 10,000) for BRCA1-A complex (excluding "regulates").

[Include "regulates"](#)

For more information, please see the [ontology relation documentation](#).

Annotations Graph Views Inferred Tree View Neighborhood Mappings

Total annotations: 456

29°C Mostly cloudy

Case Study Format

Title

Bioinformatics Analysis of the Human BRCA1 Gene Using NCBI Databases

Background

A patient with a family history of hereditary breast cancer underwent genetic sequencing. The BRCA1 gene was selected for bioinformatics analysis to understand its structure, function, and clinical significance.

Objective

To analyze the BRCA1 gene using NCBI Gene, ClinVar, BLAST, and Gene Ontology databases and understand its biological and clinical importance

Databases Used

- NCBI Gene
- NCBI Nucleotide
- NCBI Protein
- ClinVar
- BLAST
- Gene Ontology

Analysis Performed

1. Gene Identification

- Gene Name: BRCA1
- Organism: Homo sapiens
- Chromosome: 17

2. Transcript Analysis

- ✓ MANE Select Transcript identified
- ✓ Transcript accession recorded
- ✓ FASTA sequence obtained

3. Protein Analysis

- ✓ Protein accession identified
- ✓ Protein length analyzed
- ✓ Protein function reviewed

4. Genome Browser

Observed:

- ✓ Exons
- ✓ Introns
- ✓ Chromosomal location

5. ClinVar

Observed:

- ✓ Pathogenic variants
- ✓ HGVS nomenclature
- ✓ Clinical significance

6. BLAST

The BRCA1 nucleotide sequence was compared against the NCBI nucleotide database.

Result:

- 100% Query Coverage
- 100% Identity
- E-value = 0.0

Interpretation:

The uploaded sequence was confirmed as the human BRCA1 gene.

7. Gene Ontology

BRCA1 participates in:

1. Biological Process
 - DNA repair
 - Cell cycle regulation
2. Molecular Function
 - DNA binding
 - Protein binding
3. Cellular Component
 - Nucleus

Final Interpretation

The bioinformatics analysis confirms that BRCA1 is a tumor suppressor gene involved in maintaining genome stability through DNA repair mechanisms. The sequence analysis, clinical variant information, BLAST alignment, and Gene Ontology collectively demonstrate the biological importance of BRCA1 in hereditary breast and ovarian cancer research.

Conclusion

This project provided hands-on experience in using major bioinformatics databases to study a clinically important human gene. It strengthened practical skills in sequence analysis, clinical variant interpretation, functional annotation, and biological data interpretation.

7. Discussion

The BRCA1 gene analysis demonstrated how different bioinformatics databases complement each other.

NCBI Gene provided gene information, ClinVar explained disease-associated variants, BLAST confirmed sequence identity, and Gene Ontology described the biological role of BRCA1. Together, these resources provide a comprehensive understanding of the gene.

8. Conclusion

This project successfully demonstrated the complete bioinformatics workflow for analyzing the BRCA1 gene using publicly available databases.

The project improved practical skills in:

- Gene analysis
- Sequence interpretation
- Clinical variant analysis
- Functional annotation
- Sequence validation

These skills are directly applicable to clinical genomics, NGS analysis, and bioinformatics research.

9. References

National Center for Biotechnology Information (NCBI). Gene Database.

<https://www.ncbi.nlm.nih.gov/gene/>

National Center for Biotechnology Information (NCBI). Nucleotide Database.

<https://www.ncbi.nlm.nih.gov/nucleotide/>

National Center for Biotechnology Information (NCBI). Protein Database.

<https://www.ncbi.nlm.nih.gov/protein/>

ClinVar Database. <https://www.ncbi.nlm.nih.gov/clinvar/>

Basic Local Alignment Search Tool (BLAST). <https://blast.ncbi.nlm.nih.gov/>

Gene Ontology Consortium. <https://geneontology.org/>